NUMBER SYSTEM

Q 1. Find the number of zeros in 121!

(a) 32

(b) 68

(c) 42

(d) None of these

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 2. What least value must be given to * so that the number 91876*2 is divisible by 8?

(a) 1

(b) 2

(c) 3

(d) 4

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 3. The sum of three consecutive even numbers is 444 more than the average of these three numbers. Which is the smallest of these three numbers?

(a) 215

(b) 223

(c) 222

(d) 220

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 4. What is the number of zeros in 100!?

(a) 15

(b) 20

(c) 18

(d) 24

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 5. The largest integer that divides the product of any four consecutive integers is

(a) 6

(b) 12

(c) 4

(d) 24

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 6. What is the unit digit in 7^{105}?

(a) 1

(b) 5

(c) 7

(d) 9

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 7. The number of zeros in the product 5 × 10 × 25 × 40 × 50 × 55 × 65 × 125 × 80

(a) 12

(b) 8

(c) 13

(d) 9

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 8. The greatest 5-digit number exactly divisible by 8 & 9 is:

(a) 99996

(b) 99960

(c) 99990

(d) 99936

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 9. What is the least number which, when 4000 is subtracted from it, the remainder becomes divisible by 7, 11, 13?

(a) 5001

(b) 2999

(c) 1000

(d) 6303

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 10. The sum of three consecutive even numbers is always divisible by:

(a) 12

(b) 6

(c) 18

(d) 24

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 11. $5^6 - 1$ is divisible by

(a) 13

(b) 31

(c) 5

(d) None of these

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 12. What is the number of zeroes in 1000! (factorial)?

(a) 1

(b) 5

(c) 2

(d) None of these

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 13. Find the number of trailing zeroes in 120!.

(a) 12

(b) 15

(c) 16

(d) None of these

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 14. What is the number of two-digit prime numbers?

(a) 25

(b) 17

(c) 21

(d) None of these

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 15. What is the unit's digit of 2^{51}?

(a) 2

(b) 8

(c) 1

(d) 4

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 16. What is the unit's digit of 213 × 212 × 347 × 468?

(a) 4

(b) 8

(c) 6

(d) 2

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 17. The last three digits in the multiplication 12345 × 54321 will be:

(a) 745

(b) 945

(c) 865

(d) 235

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 18. Which among the following is prime?

(a) 668

(b) 558

(c) 900

(d) None of these

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 19. The prime numbers dividing 143 and leaving a remainder of 3 in each case are:

(a) 2 and 11

(b) 3 and 7

(c) 5 and 7

(d) 11 and 13

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 20. A two-digit number is k times the sum of its digits. The number formed by reversing its digits is m times the sum of its digits. Find m.

(a) 10 – k

(b) 11 – k

(c) 9 – k

(d) k – 1

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 21. If the largest three-digit number is subtracted from the smallest five-digit number, the remainder is:

(a) 9000

(b) 9001

(c) 90001

(d) 90007

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 22. The greatest four-digit number exactly divisible by 88 is:

(a) 8888

(b) 9944

(c) 9988

(d) 9998

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 23. If the number 4A306768B2 is divisible by both 8 and 11, the smallest possible values of A and B will be:

(a) A = 3, B = 5

(b) A = 5, B = 4

(c) A = 5, B = 2

(d) A = 5, B = 3

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 24. What least value must be given to * so that the number 97215*6 is divisible by 12?

(a) 3

(b) 2

(c) 1

(d) 5

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 25. Find the least value of * for which 4832*18 is divisible by 11.

(a) 100

(b) None

(c) 12

(d) 11

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 26. The number 1X38X is divisible by 12 when X = P, and by 9 when X = Q. What is P + Q?

(a) 2

(b) 3

(c) 6

(d) 9

CHITKARA UNIVERSITY

NUMBER SYSTEM

Q 27. If x and y are the two digits of the number 653xy such that the number is divisible by 80, then x + y = ?

(a) 2 or 6

(b) 4

(c) 4 or 8

(d) 8

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 1. Find LCM of 12, 15, 20

(a) 30

(b) 60

(c) 90

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 2. HCF and LCM of two numbers is 21 and 84 respectively. If the ratio of the two numbers is 1:4, the largest of the two numbers is

(a) 84

(b) 108

(c) 100

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 3. Find the largest scale size to measure accurately the three equilateral triangles with sides measuring 76 cm, 114 cm and 152 cm.

(a) 19

(b) 38

(c) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 4. Find the least number of soldiers in a regiment such that they stand in rows of 15, 20, 25 and form a perfect square

(a) 900

(b) 1600

(c) 2500

(d) 450

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 5. Sum of two numbers is 56. If their LCM is 105, then the numbers are

(a) 7, 49

(b) 22, 34

(c) 24, 32

(d) 35, 21

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 6. Find out the HCF of 3^5, 3^9 and 3^{14}

(a) 3^6

(b) 3^9

(c) 3^5

(d) 3^{14}

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 7. Find HCF of 2, 4, 8

(a) 2

(b) 4

(c) 6

(d) 8

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 8. HCF is 16, LCM is 160, one of the numbers is 32. Find the other number?

(a) 20

(b) 60

(c) 80

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 9. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.

(a) 4

(b) 7

(c) 9

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 10. Three numbers are in the ratio 1:2:3. If their HCF is 12, find the numbers?

(a) 8, 16, 24

(b) 12, 24, 36

(c) 6, 12, 18

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 11. Two numbers are in the ratio 15:11. If their HCF is 13, find the numbers?

(a) 135 and 99

(b) 165 and 121

(c) 195 and 143

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 12. Find the HCF of 2.4, 0.36 and 7.2

(a) 12

(b) 0.012

(c) 1.2

(d) 0.12

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 13. The numbers between 200 and 300, which when divided by 6, 8 or 9 leave no remainder in each case, are

(a) 300, 243

(b) 298, 256

(c) 288, 216

(d) 258, 294

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 14. The HCF and product of two numbers are 15 and 6300 respectively. The number of possible pairs of the numbers is

(a) 4

(b) 3

(c) 2

(d) 1

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 15. The greatest number that will exactly divide 300 and 120

(a) 60

(b) 40

(c) 75

(d) 45

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 16. The HCF and LCM of two numbers are 13 and 455 respectively. If one of the numbers lies between 75 and 125, then that number is:

(a) 78

(b) 91

(c) 104

(d) 117

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 17. The number between 3000 and 4000 which is exactly divisible by 30, 36 and 80 is:

(a) 3625

(b) 3250

(c) 3500

(d) 3600

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 18. The HCF of two numbers is 12 and their difference is also 12. The numbers are:

(a) 12, 84

(b) 100, 112

(c) 40, 52

(d) 84, 96

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 19. The LCM of two numbers is 2376 while their HCF is 33. If one number is 297, the other number is:

(a) 216

(b) 264

(c) 642

(d) 792

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 20. Find the HCF of 4, 8, 16, 32 and 15.

(a) 4

(b) 8

(c) 1

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 21. The product of two numbers is 1280 and their HCF is 8. The LCM of the numbers is:

(a) 140

(b) 160

(c) 183

(d) 182

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 22. Three numbers are in the ratio 2:3:4 and their HCF is 12. The LCM of the numbers is:

(a) 144

(b) 192

(c) 96

(d) 72

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 23. Find the LCM of 22, 540 and 108.

(a) 5940

(b) 5640

(c) 4740

(d) 4940

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 24. The LCM of two numbers is 1296 and their HCF is 96. If one number is 864, the other number is:

(a) 72

(b) 64

(c) 144

(d) 1164

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 25. The LCM of two numbers is 1920 and their HCF is 16. If one number is 128, find the other number.

(a) 204

(b) 240

(c) 360

(d) 320

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 26. Three sets of English, Mathematics and Science books contain 336, 240 and 96 books respectively. They must be stacked subject-wise, with the height of every stack the same. The total number of stacks will be:

(a) 14

(b) 21

(c) 22

(d) 48

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 27. Find the LCM of 16, 24 and 36.

(a) 123

(b) 111

(c) 11

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 28. Find the least number of soldiers in a regiment such that they can stand in rows of 18, 15 and 25 and also form a perfect square.

(a) 900

(b) 1600

(c) 2500

(d) 450

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 29. Find the least perfect square divisible by 8, 9 and 10.

(a) 4000

(b) 3600

(c) 6400

(d) 111

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 30. Find the greatest number that will exactly divide 200 and 320.

(a) 30

(b) 40

(c) 50

(d) 45

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 31. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together, including the one at the beginning?

(a) 16

(b) 15

(c) 14

(d) None of these

CHITKARA UNIVERSITY

H.C.F. & L.C.M.

Q 32. What is the least number which, when doubled, becomes exactly divisible by 9, 15, 21 and 30?

(a) 196

(b) 189

(c) 630

(d) 315

CHITKARA UNIVERSITY

AVERAGE

Q 1. The mean temperature of Monday to Wednesday was 37°C and of Tuesday to Thursday was 34°C. If the temperature on Thursday was 4/5 that of Monday, the temperature on Thursday was:

(a) 10

(b) 12

(c) 11

(d) None of These

CHITKARA UNIVERSITY

AVERAGE

Q 2. I went to Delhi at the speed of 200 km/hr but suddenly I returned to the same place at the speed of 600 km/hr. What is my average speed?

(a) 300 km/hr

(b) 400 km/hr

(c) 366.66 km/hr

(d) None of These

CHITKARA UNIVERSITY

AVERAGE

Q 3. The average weight of 11 players of Indian cricket team is increased by 1 kg, when one player of the team weighing 55 kg is replaced by a new player. The weight of the new player is

(a) 55

(b) 64

(c) 66

(d) None of These

CHITKARA UNIVERSITY

AVERAGE

Q 4. Average of 7 consecutive odd numbers if the smallest of those numbers is denoted by k

(a) k+4

(b) k+7

(c) k+6

(d) 7k

CHITKARA UNIVERSITY

AVERAGE

Q 5. The average age of 30 students of a class is 30 years. When the age of the class teacher is also included, the average age of the whole class increases by 1 year. The age of the class teacher is:

(a) 31 years

(b) 60 years

(c) 61 years

(d) None of these

CHITKARA UNIVERSITY

AVERAGE

Q 6. There were five sections in MAT paper. The average score of Pooja in first 3 sections was 83 and the average in the last 3 sections was 97 and the average of all the sections (i.e. whole paper) was 92, then her score in the third section was:

(a) 85

(b) 92

(c) 88

(d) None of these

CHITKARA UNIVERSITY

AVERAGE

Q 7. Find the average of the squares of the first 111 natural numbers

(a) 300

(b) 400

(c) 500

(d) None

CHITKARA UNIVERSITY

AVERAGE

Q 8. The average of five numbers is 306. The average of the first two numbers is 431 and the average of the last two numbers is 214. What is the third number?

(a) 0

(b) None of these

(c) 3

(d) 1

CHITKARA UNIVERSITY

AVERAGE

Q 9. In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs?

(a) 6.25

(b) 6.5

(c) 6.75

(d) 7

CHITKARA UNIVERSITY

AVERAGE

Q 10. The average of first 50 odd natural numbers is:

(a) 50

(b) 55

(c) 51

(d) 101

CHITKARA UNIVERSITY

AVERAGE

Q 11. Find the average of the first 1115 natural numbers?

(a) 431

(b) 558

(c) 430

(d) 403

CHITKARA UNIVERSITY

AVERAGE

Q 12. Three types of rice whose rates are Rs 38, Rs 43 and Rs 49 per kg are blended together to make 15 kg of new blend in which there are 8 kg, 4 kg, 3 kg of the respective types of rice. The average price of the new blend of rice is

(a) Rs 41.53

(b) Rs 43

(c) Rs 40

(d) Rs 43.3

CHITKARA UNIVERSITY

AVERAGE

Q 13. The average length of first 3 fingers is 3 inches and the average length of the other 2 fingers i.e. thumb and index finger is 2.8 inches. If the length of the index finger is 3 inches then the length of the thumb is:

(a) 2 inches

(b) 2.6 inches

(c) 3 inches

(d) None of these

CHITKARA UNIVERSITY

AVERAGE

Q 14. The average of first 99 even numbers is:

(a) 9999

(b) 100

(c) 9801

(d) 9009

CHITKARA UNIVERSITY

AVERAGE

Q 15. Find the average of natural numbers from 1 to 67?

(a) 33.5

(b) 34

(c) 50

(d) 67

CHITKARA UNIVERSITY

AVERAGE

Q 16. The average of first 100 natural numbers is

(a) 100

(b) 50

(c) 50.5

(d) 55

CHITKARA UNIVERSITY

AVERAGE

Q 17. The cost of the Red, Green and Blue colours per kg is Rs. 20, Rs. 15 and Rs. 18 respectively. Rang Mahal is a renowned building in which these three colours are being used in the ratio of 3 : 2 : 4. The average cost of all the three colours used per kg is:

(a) 18

(b) 20

(c) 17.66

(d) Can't be determined

CHITKARA UNIVERSITY

AVERAGE

Q 18. Find the average of all the numbers between 6 and 34 that are divisible by 5.

(a) 18

(b) 19

(c) 20

(d) 22

CHITKARA UNIVERSITY

AVERAGE

Q 19. The average of 5 terms is 50. If the first four terms are 45, 42, 119 and 84, what is the last term?

(a) 56

(b) −20

(c) −40

(d) −50

CHITKARA UNIVERSITY

AVERAGE

Q 20. If the average of 8 terms is 40 and the average of the first 6 terms is 35, what is the average of the remaining 2 terms?

(a) 30

(b) 55

(c) 40

(d) 42

CHITKARA UNIVERSITY

AVERAGE

Q 21. Find the average of the first 101 natural numbers.

(a) 47

(b) 48

(c) 51

(d) 49

CHITKARA UNIVERSITY

AVERAGE

Q 22. Find the average of the first 7 natural numbers.

(a) 4

(b) 6

(c) 7

(d) 9

CHITKARA UNIVERSITY

AVERAGE

Q 23. The average of three consecutive even numbers is M. If the next five even numbers are also included, the average of all 8 numbers will be:

(a) M + 6

(b) M + 3

(c) M + 5

(d) M + 4

CHITKARA UNIVERSITY

AVERAGE

Q 24. The average of 7 consecutive positive integers is 10. The average of the lowest and highest numbers is:

(a) 7

(b) 10

(c) 15

(d) Data insufficient

CHITKARA UNIVERSITY

AVERAGE

Q 25. In a class, there are 20 boys. Their average age decreases by 2 months when a boy aged 18 years is replaced by a new boy. Find the age of the new boy.

(a) 16 years

(b) 21 years 4 months

(c) 14 years 8 months

(d) 15 years

CHITKARA UNIVERSITY

AVERAGE

Q 26. The average weight of 20 students is 45 kg. A student weighing 40 kg replaces an old student, and the class average decreases by 1 kg. The weight of the replaced student is:

(a) 55 kg

(b) 50 kg

(c) 60 kg

(d) None of these

CHITKARA UNIVERSITY

AVERAGE

Q 27. The average price of 3 diamonds of the same weight is Rs. 5 crore. The average price of the two costliest diamonds is double the price of the cheapest diamond. The price of the cheapest diamond is:

(a) Rs. 3 crore

(b) Rs. 5 crore

(c) Rs. 1.66 crore

(d) Cannot be determined

CHITKARA UNIVERSITY

AVERAGE

Q 28. Kamlesh bought 65 books for Rs. 1050 from one shop and 50 books for Rs. 1020 from another. What was the average price per book?

(a) Rs. 36.4

(b) Rs. 18.2

(c) Rs. 24

(d) Rs. 18

CHITKARA UNIVERSITY

AVERAGE

Q 29. Pankaj travelled to the post office at 60 km/h. On the return journey, he covered half the distance at 10 km/h and the remaining half at 30 km/h. The average speed for the whole journey was:

(a) 5.67 km/h

(b) 24 km/h

(c) 22.88 km/h

(d) 5.45 km/h

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 1. Find the greatest number of five digits which when divided by 3, 5, 8, 12 have 2 as a remainder

(a) 99999

(b) 99958

(c) 99960

(d) 99962

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 2. The remainder obtained when a prime number greater than 6 is divided by 6 is

(a) 1 or 3

(b) 1 or 5

(c) 3 or 5

(d) 4 or 5

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 3. When 'n' is divided by 5 the remainder is 2. What is the remainder when n^2 is divided by 5?

(a) 2

(b) 3

(c) 1

(d) 4

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 4. Find the smallest number which leaves the remainders 13, 41 and 29 at the end when divided by 20, 48 and 36 respectively.

(a) 187

(b) 713

(c) 720

(d) 727

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 5. A number when divided successively by 4 and 5 leaves remainder 1 and 4 respectively. When it is successively divided by 5 and 4 the respective remainders will be

(a) 4, 1

(b) 3, 2

(c) 2, 3

(d) 1, 2

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 6. The remainder when 7^{84} is divided by 342 is

(a) 0

(b) 1

(c) 49

(d) 341

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 7. When a number is divided by 357 the remainder is 39. If that number is divided by 17, the remainder will be:

(a) 0

(b) 5

(c) 11

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 8. The least number, which when divided by 12, 15, 20 or 54 leaves a remainder of 4 in each case is:

(a) 450

(b) 454

(c) 540

(d) 544

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 9. The least number which when divided by 16, 18, 20 and 25 leaves 4 as a remainder in each case but when divided by 7 leaves no remainder is

(a) 17004

(b) 18000

(c) 18002

(d) 18004

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 10. A number, when divided by 136, leaves remainder 36. If the same number is divided by 17, the remainder will be

(a) 9

(b) 3

(c) 2

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 11. What is the remainder when 4^{1000} is divided by 7?

(a) 6

(b) 2

(c) 4

(d) None of these

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 12. What is the remainder when 6729 is divided by 35?

(a) 11

(b) 7

(c) 9

(d) 13

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 13. If $71^{83} + 73^{83}$ is divided by 36, the remainder is:

(a) 13

(b) 9

(c) 0

(d) 8

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 14. Find the least number which when divided by 36, 48 and 64 leaves the remainders 25, 37 and 53 respectively.

(a) 565

(b) 560

(c) 555

(d) 550

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 15. The least number which when divided by 5, 6, 7 and 8 leaves the remainder 3, but when divided by 9 leaves no remainder, is:

(a) 1766

(b) 1683

(c) 2327

(d) 1895

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 16. What is the remainder when 2^{100} is divided by 3?

(a) 3

(b) 0

(c) 1

(d) 2

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 17. What is the remainder when 2^{100} is divided by 7?

(a) 1

(b) 2

(c) 3

(d) 4

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 18. If 17^{200} is divided by 18, the remainder is:

(a) 17

(b) 16

(c) 1

(d) 2

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 19. What is the remainder when 293 is divided by 7?

(a) 1

(b) 2

(c) 4

(d) 6

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 20. A number, when divided by 357, leaves a remainder of 39. If the number is divided by 7, the remainder will be:

(a) 4

(b) 9

(c) 5

(d) 11

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 21. A number, when divided by 221, leaves a remainder of 64. What is the remainder when the same number is divided by 13?

(a) 0

(b) 1

(c) 11

(d) 12

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 22. A number, when divided by 296, gives a remainder of 75. When the same number is divided by 37, the remainder will be:

(a) 1

(b) 2

(c) 8

(d) 11

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 23. The integers 34041 and 32506, when divided by a three-digit integer n, leave the same remainder. What is n?

(a) 289

(b) 367

(c) 453

(d) 307

CHITKARA UNIVERSITY

REMAINDER THEOREM

Q 24. Find the least number which, when divided by 27, 35, 45 and 49, leaves a remainder of 6 in each case.

(a) 5523

(b) 6620

(c) 6623

(d) 6621

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NUMBER SYSTEM - ANSWER KEY

Q 1. d) None of these

Q 2. c) 3

Q 3. d) 220

Q 4. d) 24

Q 5. d) 24

Q 6. c) 7

Q 7. d) 9

Q 8. d) 99936

Q 9. a) 5001

Q 10. b) 6

Q 11. b) 31

Q 12. d) None of these

Q 13. d) None of these

Q 14. c) 21

Q 15. b) 8

Q 16. c) 6

Q 17. a) 745

Q 18. d) None of these

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NUMBER SYSTEM - ANSWER KEY (Contd.)

Q 19. c) 5 and 7

Q 20. b) 11 − k

Q 21. b) 9001

Q 22. b) 9944

Q 23. d) A = 5, B = 3

Q 24. a) 3

Q 25. b) None

Q 26. b) 3

Q 27. a) 2 or 6

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NUMBER SYSTEM - NOTES & REVIEW ITEMS

Q 12. Source note: 1000! has 249 trailing zeroes, so "None of these" is correct.

Q 13. Source note: 120! has 28 trailing zeroes, so "None of these" is correct.

Q 25. Source note: the digit that makes 4832*18 divisible by 11 is 7, but 7 is not among the listed options; the source marks "None" as the best match. [Review]

Q 27. Source note: option (a) is transcribed as "2 or 6" although the source says the computed value is 2. [Review: 653xy is divisible by 80 only for 65360, i.e. x = 6, y = 0, so x + y = 6. Option (a) still contains the right value; the question is kept exactly as supplied.]

H.C.F. & L.C.M. - ANSWER KEY

Q 1. b) 60

Q 2. a) 84

Q 3. b) 38

Q 4. a) 900

Q 5. d) 35, 21

Q 6. c) 3^5

Q 7. a) 2

Q 8. c) 80

Q 9. a) 4

Q 10. b) 12, 24, 36

Q 11. c) 195 and 143

Q 12. d) 0.12

Q 13. c) 288, 216

Q 14. c) 2

Q 15. a) 60

Q 16. b) 91

Q 17. d) 3600

Q 18. d) 84, 96

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H.C.F. & L.C.M. - ANSWER KEY (Contd.)

Q 19. b) 264

Q 20. c) 1

Q 21. b) 160

Q 22. a) 144

Q 23. a) 5940

Q 24. c) 144

Q 25. b) 240

Q 26. a) 14

Q 27. d) None of these

Q 28. a) 900

Q 29. b) 3600

Q 30. b) 40

Q 31. a) 16

Q 32. d) 315

H.C.F. & L.C.M. - NOTES & REVIEW ITEMS

Q 24. [Review] The data is inconsistent: HCF 96 does not divide LCM 1296, and 864 does not divide 1296. The supplied answer (c) 144 follows from HCF x LCM = product of the numbers (1296 x 96 / 864 = 144). Kept as supplied.

Q 27. Source note: the LCM of 16, 24 and 36 is 144, so "None of these" is correct.

Q 28. [Info] Similar to original Q 4 (rows of 15, 20, 25) but with rows of 18, 15 and 25; retained as a different question.

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AVERAGE - ANSWER KEY

Q 1. d) None of These

Q 2. a) 300 km/hr

Q 3. c) 66

Q 4. c) k+6

Q 5. c) 61 years

Q 6. d) None of these

Q 7. d) None

Q 8. b) None of these

Q 9. a) 6.25

Q 10. a) 50

Q 11. b) 558

Q 12. a) Rs 41.53

Q 13. b) 2.6 inches

Q 14. b) 100

Q 15. b) 34

Q 16. c) 50.5

Q 17. a) 18

Q 18. c) 20

AVERAGE - ANSWER KEY (Contd.)

Q 19. c) −40

Q 20. b) 55

Q 21. c) 51

Q 22. a) 4

Q 23. c) M + 5

Q 24. b) 10

Q 25. c) 14 years 8 months

Q 26. c) 60 kg

Q 27. a) Rs. 3 crore

Q 28. d) Rs. 18

Q 29. b) 24 km/h

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REMAINDER THEOREM - ANSWER KEY

Q 1. d) 99962

Q 2. b) 1 or 5

Q 3. d) 4

Q 4. b) 713

Q 5. c) 2, 3

Q 6. b) 1

Q 7. b) 5

Q 8. d) 544

Q 9. d) 18004

Q 10. c) 2

Q 11. c) 4

Q 12. c) 9

Q 13. c) 0

Q 14. a) 565

Q 15. b) 1683

Q 16. c) 1

Q 17. b) 2

Q 18. c) 1

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REMAINDER THEOREM - ANSWER KEY (Contd.)

Q 19. d) 6

Q 20. a) 4

Q 21. d) 12

Q 22. a) 1

Q 23. d) 307

Q 24. d) 6621

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REMAINDER THEOREM - NOTES & REVIEW ITEMS

Q 19. [Review] Printed as "293" in the source. If 2^{93} was intended, the remainder would be 1, not 6. As printed, 293 divided by 7 leaves 6, which matches the supplied answer (d).

Skipped. TXT Remainder Q4 (remainder of 4^{1000} divided by 7) is already Q 11 of the original deck, so it was not added again. Its options differ slightly: option (b) is 5 in the TXT but 2 in the original; the correct answer (4) agrees.

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